Triage Report — WattAgents

1. Submission Summary

Brief description of what the team is proposing (keep these explicitly to what has been selected)

- **Industry:** Utilities Sector
- **Use Case:** Real-time resilience Approach — Operations Assistant. The submission maps to the Utilities scenario of control-room resilience, using a multi-agent system to monitor reactor telemetry, detect compound safety risks, and trigger shutdown / evacuation decisions.
- **Track: Warn — not clearly declared.** Based on the stack, this appears closest to **Track A (GitHub + Azure + optional Microsoft Fabric)** because the architecture is primarily Azure + custom code. The mention of Copilot Studio / Foundry on one slide is inconsistent with the rest of the described implementation.
- **Stage:** Submission Review

Cross-cutting checks

Check	Status	Notes
Environment Track declared	**Warn**	The submission does not explicitly declare A / B / C / D. The implementation mostly aligns to **Track A** through Azure OpenAI, Event Hubs, Cosmos DB, AI Search, AKS, FastAPI, and React. However, one architecture slide also lists Copilot Studio and Foundry without explaining their role, creating track ambiguity.
Industry Use Case aligned	**Pass**	Strong alignment to **Utilities Sector — Real-time resilience Approach**. The proposal addresses control-room supervision, real-time risk detection, and rapid response in a critical infrastructure setting. While framed around a nuclear incident rather than power grid operations, it is still materially aligned to resilient utility operations.
Azure WAF aligned	**Warn**	Reliability, Performance Efficiency, and Cost Optimisation are addressed reasonably well through fallback logic, timeout handling, selective LLM use, and cost-per-run estimates. Security and Operational Excellence are only partially covered; identity model, deployment operations, and monitoring detail are limited.
Azure WAF AI Pillar aligned	**Warn**	Agent design and grounding are partially addressed via structured JSON outputs, guardrails, and AI Search over IAEA docs. Responsible AI is discussed mainly as anti-hallucination and non-override rules, but broader governance, evaluation, model lifecycle, and AI observability are not sufficiently detailed.
Power Platform WAF aligned	**Fail**	Copilot Studio is named, but no Power Platform / Copilot Studio architecture, connectors, environments, governance, or UX flow is actually described. This looks incidental rather than implemented.
Microsoft Security & Monitoring	**Warn**	Some security thinking is present: environment variables for secrets, hard guardrails, and limited logging of sensitive telemetry. But Microsoft security and monitoring controls are not concretely defined: no Entra ID, RBAC, Defender, Monitor, Log Analytics, alerting, or audit strategy is described.

2. Criteria Scoring

Architecture

Dimension	Score	Justification
Clarity & Flow	**8/10**	The end-to-end design is one of the strongest parts of the submission. The team clearly describes a deterministic pipeline from Sensor → Risk → Decision → Evacuation → Comms → Dashboard, with named agents and a readable 7-layer architecture. This scores well on **Clear Architecture Flow**, though some inconsistency between slides around Foundry / Copilot Studio slightly reduces clarity.
Platform Fit	**7/10**	Azure Event Hubs, Azure OpenAI, Cosmos DB, AI Search, FastAPI, and AKS are generally appropriate for a telemetry-driven, agentic control-room scenario, satisfying **Platform Fit** well. However, the track is not explicitly declared, and the unexplained inclusion of Copilot Studio / Foundry / ADLS weakens confidence that the platform choices are fully intentional and aligned to one track.
Data / Governance / Security	**6/10**	The submission gives useful detail on typed reactor state, JSON schema-constrained outputs, and state history in Cosmos DB, which supports **Data & Governance**. Security is only partially addressed: secret handling through environment variables and hard guardrails are good starts, but there is limited coverage of identity, role-based access, data classification, retention, auditability, compliance boundaries, or Microsoft-native security controls.
Reliability / Performance / Cost	**8/10**	Strong evidence is provided for **Reliability**, **Performance**, and **Cost Awareness**. The team describes 5-second LLM timeouts, fallback to deterministic logic, limited LLM invocation points, live latency tracking, and a low estimated run cost. This is a well-considered architecture dimension, though production SLOs, resilience testing, and capacity assumptions are still missing.
Scalability / Integration / Provisioning	**7/10**	The proposal addresses **Scalability** through decoupled agents, pub/sub patterns, and AKS/Event Hubs scalability claims. **Integration** is reasonably described through SDK clients, SSE endpoint, and orchestrator patterns. **Provisioning Readiness** is only moderate: services are listed and smoke tests are mentioned, but there is no explicit IaC, deployment pipeline, environment topology, or provisioning checklist.

Architecture Subtotal: 36/50

Use Case

Dimension	Score	Justification
Value & KPI Impact	**9/10**	This is a high-significance problem with clear stakeholder value, strongly meeting **Problem Significance**, **User Impact**, and **KPI Impact**. The submission provides concrete metrics such as earlier shutdown, earlier evacuation, and low simulation cost. The main limitation is that the impact is demonstrated through historical replay/simulation rather than validated real-world utility operations.
Innovation	**8/10**	The use of a multi-agent safety architecture replaying INSAG-7 events with blended rule-based and LLM scoring is creative and differentiated, scoring well on **Innovation**. The approach goes beyond a generic chatbot by combining telemetry monitoring, decision ladders, evacuation logic, and plume communications. It stops short of a 9–10 because some elements still resemble established agentic monitoring patterns rather than a wholly novel Microsoft-specific pattern.

AI Fit	**7/10**	AI is reasonably justified because the solution aims to detect compound failure modes and support rapid high-stakes decisions, which aligns with **AI Fit**. The submission also addresses **Responsible Design** through non-negotiable guardrails, anti-hallucination logic, and constrained outputs. However, in such a safety-critical domain, the rationale for where AI is necessary versus where deterministic control should dominate needs deeper treatment, especially around human oversight, explainability, and failure consequences.
UX Simplicity	**6/10**	The live dashboard and decision ladder suggest a usable operator experience, but the submission gives limited detail on **User Experience**. There is little about operator workflows, alert design, approval paths, escalation UI, or how humans interpret and trust recommendations under pressure. The concept is understandable, but the UX remains under-specified.
Build–Scale–Reuse	**8/10**	The team makes a credible case for **Feasibility** with a Python implementation, smoke-test mode, and a bounded simulation use case. **Scalability & Security** are partially supported by Azure-native services. **Reusability (MCP)** is a notable strength: the agent modules and guardrail-first pattern appear reusable across oil & gas, aviation, and pharma. The score is reduced by the lack of explicit MCP packaging or reusable asset definition.

Use Case Subtotal: 38/50

Total

Total Score: 74/100

3. Strengths

- Strong **Value & KPI Impact**: the submission clearly frames a high-stakes utility resilience problem and quantifies expected outcomes with specific KPIs such as shutdown lead time and evacuation lead time.
- Strong **Clarity & Flow**: the 5-agent / 7-layer architecture is easy to follow, with clear roles, data flow, and decision sequencing.
- Strong **Reliability / Performance / Cost**: deterministic fallback logic, timeout handling, selective LLM invocation, and cost-per-run estimation show mature architectural thinking.
- Good **Innovation**: the blend of hard guardrails plus advisory LLM reasoning for compound-risk detection is a thoughtful and differentiated design choice.
- Good **Build–Scale–Reuse**: the architecture has credible reuse potential across other safety-critical industries, which aligns well with the hackathon objective of producing reusable IP.

4. Gaps & Risks

- **Environment Track** is not explicitly declared, which creates ambiguity against the hackathon requirement to align the architecture to Track A / B / C / D.
- **Platform Fit** is weakened by inconsistent references to Copilot Studio, Foundry, GPT-4.1-mini, and ADLS that are not integrated into the main architecture narrative.
- **Data / Governance / Security** is incomplete for a safety-critical scenario: identity, RBAC, audit trails, retention, regulatory controls, and Microsoft-native security services are not sufficiently defined.
- **Azure WAF AI Pillar** coverage is partial: responsible AI is discussed narrowly as anti-hallucination and hard thresholds, but model lifecycle, evaluation, observability, and human oversight are not well developed.
- **UX Simplicity** is under-detailed: the dashboard exists, but the operator journey, intervention points, and alert usability under stress are not described.
- The **Industry Use Case alignment** is good but not perfect: the hackathon's Utilities example emphasizes modern intelligent grid / control-room resilience, while this submission focuses on a historical nuclear simulation. The mapping works, but it should be made more explicit.

- **Provisioning Readiness** is only moderate because there is no clear IaC or deployment model, despite listing services and a smoke-test path.

5. Triage Verdict

Good — 74/100

6. Top Actionable Improvements

- Explicitly declare the intended **environment track** and remove or justify any out-of-track technologies; this will materially improve **Platform Fit** and the **Environment Track declared** cross-check.
- Strengthen **Data / Governance / Security** by adding a concrete Microsoft security model: Entra ID/RBAC, managed identities, Key Vault, logging boundaries, data retention, audit trails, and monitoring/alerting with Azure Monitor / Log Analytics.
- Expand the **Azure WAF AI Pillar** treatment with clear responsible AI controls: explainability of risk scores, human-in-the-loop escalation policy, model evaluation criteria, prompt/version management, and AI observability.
- Make the **UX Simplicity** more concrete by showing the operator workflow: what appears on the dashboard, which actions are automated vs approved by humans, how alerts are prioritised, and how emergency decisions are explained.
- Improve **Scalability / Integration / Provisioning** by adding a simple deployment blueprint or IaC view showing environments, services, networking, integration points, and how the solution would be reproducibly provisioned in Azure.

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